

HDOC-31

Aim:

To write a **menu-driven C program** using a 2D array and switch-case to perform the following matrix operations:

1. Find the **sum of all elements** of the matrix.
2. Find the **sum of each row** and store the results in a 1D array.
3. Find and display the **transpose** of the matrix.
4. Exit the program.

Task1: Algorithm:

Main Algorithm

1. Start.
2. Declare variables rows, cols, i, j, choice, sum and an array rowSum.
3. Input the number of rows and columns.
4. Declare a 2D array matrix[rows][cols].
5. Input the elements of the matrix using nested for loops.
6. Display the following menu:
 - 1 – Sum of all elements.
 - 2 – Sum of each row.
 - 3 – Transpose of matrix.
 - 4 – Exit.
7. Read the user's choice.
8. Use a switch statement to perform the selected operation.

Case 1: Sum of All Elements

9. Initialize sum = 0.
10. Traverse all elements of the matrix using nested loops.
11. Add each element to sum.
12. Display sum.

Case 2: Sum of Each Row

13. For each row, initialize rowSum[i] = 0.
14. Traverse all columns of that row.
15. Add each element to rowSum[i].
16. Display the sum of each row.
17. Display the complete rowSum array.

Case 3: Transpose of Matrix

18. Traverse the matrix with the column loop as the outer loop and the row loop as the inner loop.
19. Display matrix[i][j] so that rows become columns.
20. Display the transposed matrix.

Case 4: Exit

21. Display "Exiting program...".
22. If the choice is not between 1 and 4, display "Invalid choice".
23. Repeat the menu until the user enters 4.
24. Stop.

Task2: Test case table:

Test Case	Matrix	Choice	Expected Output
1	1 2 3 / 4 5 6	1	Sum = 21
2	1 2 3 / 4 5 6	2	Row sums = [6, 15]
3	1 2 3 / 4 5 6	3	Transpose = 1 4 / 2 5 / 3 6
4	5	1	Sum = 5
5	Any valid matrix	4	Exiting program...

Task3: C Program:

```
#include <stdio.h>
int main()
{
    int rows, cols;
    int i, j, choice;
    int sum, rowSum[100];

    printf("Enter the number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

    int matrix[rows][cols];

    printf("\nEnter the matrix elements:\n");

    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
```

```

    {
        scanf("%d", &matrix[i][j]);
    }
}

do
{
    printf("\n--- MENU ---\n");
    printf("1. Find sum of all elements\n");
    printf("2. Find sum of each row\n");
    printf("3. Find transpose of matrix\n");
    printf("4. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            sum = 0;

            for(i = 0; i < rows; i++)
            {
                for(j = 0; j < cols; j++)
                {
                    sum = sum + matrix[i][j];
                }
            }

            printf("Sum of all elements = %d\n", sum);
            break;

        case 2:
            printf("\nRow sums:\n");

            for(i = 0; i < rows; i++)
            {
                rowSum[i] = 0;

                for(j = 0; j < cols; j++)
                {
                    rowSum[i] = rowSum[i] + matrix[i][j];
                }

                printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
            }

```

```

printf("\nThe row-sum array contains: [");

for(i = 0; i < rows; i++)
{
    printf("%d", rowSum[i]);

    if(i < rows - 1)
    {
        printf(", ");
    }
}

printf("]\n");
break;

case 3:
    printf("\nTranspose of the matrix:\n");

    for(j = 0; j < cols; j++)
    {
        for(i = 0; i < rows; i++)
        {
            printf("%d\t", matrix[i][j]);
        }

        printf("\n");
    }
    break;

case 4:
    printf("Exiting program...\n");
    break;

default:
    printf("Invalid choice! Please enter 1 to 4.\n");
}

} while(choice != 4);

return 0;
}

```

Task4: Output:

```
(kali@kali)~/hdcc31.c
$ cc ques1.c

(kali@kali)~/hdcc31.c
$ ./a.out
Enter the number of rows and columns: 2 3

Enter the matrix elements:
1
2
3
4
5
6

---MENU---
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit
Enter your choice: 1
Sum of all elements= 21

---MENU---
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit
Enter your choice: 2

Row sums:
Sum of row 1 = 6
Sum of row 2 = 15

The row-sum array contains: [6,15]

---MENU---
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit
Enter your choice: 3

Transpose of the matrix:
1      4
2      5
3      6
```

```
(kali@kali)~/hdcc31.c
$ ./a.out
Enter the number of rows and columns: 1
1

Enter the matrix elements:
5

---MENU---
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit
Enter your choice: 1
Sum of all elements= 5

---MENU---
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit
Enter your choice: 4
Exiting program...
```